

GoBench: Evaluating LLMs on the Game of Go

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Abstract

We introduce GoBench, which evaluates LLMs on full games of 9×9 Go and uses KataGo players as Elo anchors. Without tools, the strongest LLM (GPT-6 Astra) reaches 2500 Elo, while the strongest KataGo player reaches 4400 Elo. By giving GPT-6 Astra 2 hours of continual learning with coding and evaluation with coding as well, it reaches 3600 Elo. We aim to keep the GoBench leaderboard up to date as long as it is not saturated: <https://rolandgao.com/blog/gobench/>

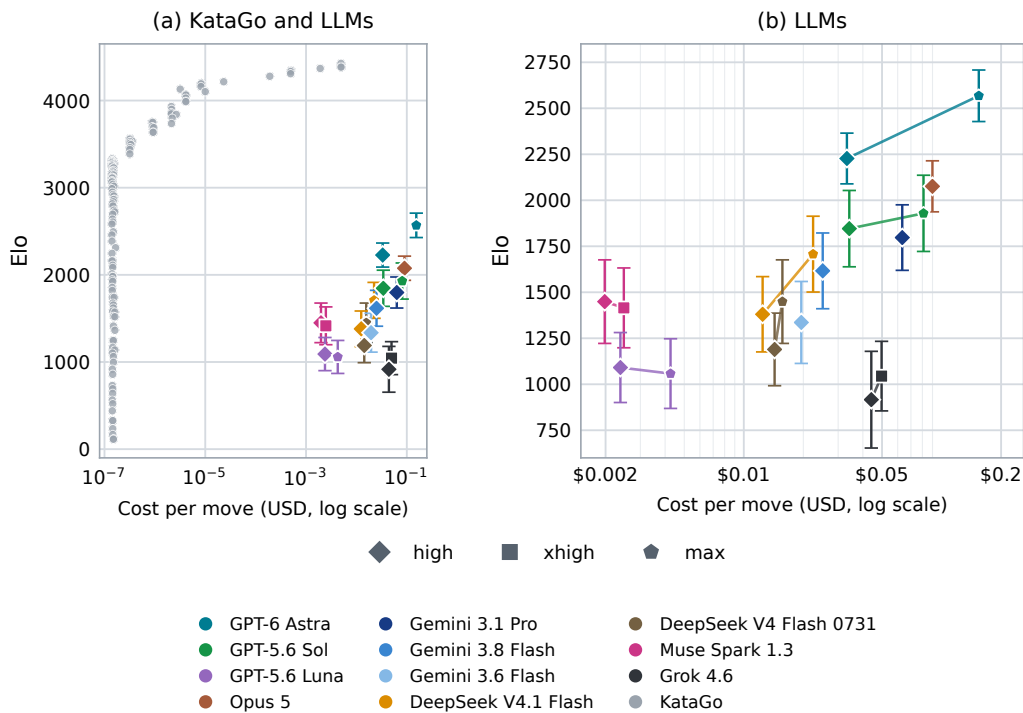


Figure 1: Elo versus cost per move. (a) 132 KataGo players and LLMs without tools. (b) Detail of LLMs. Error bars show 95% confidence intervals.

1 Introduction

Current models display “jagged intelligence” [Karpathy, 2024] as they approach top human performance in math and coding but lag in other domains. On our path toward AGI, we want to develop AI systems that can learn new domains at lower cost and with less human supervision.

The game of Go serves as a new domain, and we can evaluate LLMs on Go in a few tracks:

1. direct evaluation with no tools
2. continual learning with coding tools, evaluation with coding tools
3. continual learning with coding tools, evaluation without tools
4. continual learning with coding tools and KataGo, evaluation without tools

Humans with no prior Go experience fall into Track 1, while humans with prior Go experience fall into Track 4. In this work, we focus on Tracks 1 and 2, leaving 3 and 4 to future work, as they require weight updates.

Track 1 measures LLMs’ general reasoning ability, as they have not been trained extensively on Go. As we will show in Section 3.1, Track 1 Elos strongly correlate with ARC-AGI 1/2 scores. GoBench remains far from saturated and can measure general reasoning beyond ARC-AGI [Chollet, 2019, Chollet et al., 2025, ARC Prize Foundation, 2026b].

Track 2 measures LLMs’ coding ability and context-based continual learning abilities. We let the LLMs perform continual learning for $\{0, 1, 2, 4, 8\}$ hours before doing the evaluation. This can help us understand the cost of learning a new domain.

We never allow LLMs to call KataGo as a tool during evaluation, as that would defeat the purpose of this evaluation.

2 Implementation overview

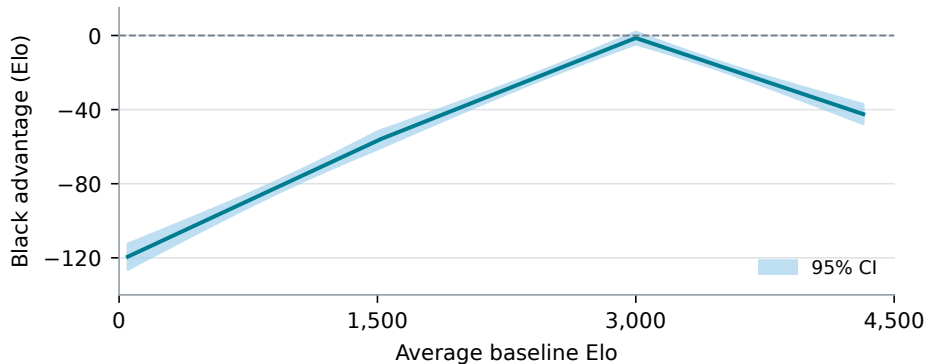


Figure 2: Black advantage in the current Elo fit, with its pointwise 95% confidence band. With the 7.0-point komi, Black is at a disadvantage at low Elos, approaches parity at 3000 Elo, and has a slight disadvantage near 4300 Elo.

We provide an implementation overview here with details in the Appendix.

We use the Bradley–Terry model [Bradley and Terry, 1952] with a Gaussian prior for Elo ratings and confidence intervals. A relative Elo difference of 400 corresponds to 10:1 odds of winning, and a difference of 800 corresponds to 100:1 odds. The random player has a fixed Elo of 0, and all other players’ Elos are derived from it.

In Go, the Black player goes first and has an advantage. We mitigate the effect of color by

1. having a second game with colors reversed for each selected matchup,
2. giving the White player 7.0 extra points during scoring,

3. and predicting the color advantage with a piecewise linear function and incorporating it into the Elo model.

The KataGo players play 320,000 games among themselves to solidify the Elo ladder. Then, each LLM plays 14 to 30 games to achieve confidence intervals of about ± 200 .

KataGo’s temperature decays exponentially from 0.5 to 0.1 to provide some variability in the opening moves. KataGo players resign after three consecutive evaluations below -0.95 . The random player never resigns and samples uniformly from the currently legal empty intersections plus pass.

Most KataGo players use `numPlayouts = 1`, which simply samples from the policy network without any tree search. For the top four katal networks, we also add variants with `numPlayouts = 60` and `numPlayouts = 600`. The strongest KataGo player in our pool is probably superhuman, but the precise Elos of top human professionals are unknown.

We use 9×9 Go boards to get high signal with only a quarter of the cost and time of a 19×19 board. According to the author of KataGo, the 9×9 board is far from solved [Wu, 2023]; the optimal moves at many positions are still unknown. We will explore larger board sizes such as 13×13 and 19×19 once LLMs start saturating the 9×9 board.

KataGo timings use a Google Cloud N1 instance (n1-custom-1-6656) with one vCPU, priced at \$0.071 per hour. For the eight players that use tree search, we estimate time per move by multiplying the measured single-playout time by the number of playouts.

In the Track 1 API comparison, LLMs use multi-turn reasoning, where reasoning tokens from prior turns are preserved. This reduces cost while maintaining Elo. Conversations reset between games or when the context reaches 250K tokens within a game. Our turn prompt contains complete information, so a conversation reset within a game is acceptable. We do not use autocompaction because it is not natively supported by all APIs. If an LLM outputs an illegal move, it is prompted again with the same turn prompt until it succeeds.

Prompt 1: GoBench evaluation prompt.

```
You are playing Black in an ongoing 9x9 Go game against another bot. It is Black to move.

Rules: White komi 7.0, positional superko, passing allowed, self-capture of stones legal.
Resolve captures by first removing opposing groups with no liberties, then removing any
friendly groups with no liberties. Self-capture remains subject to positional superko.
Resigning loses immediately. Two consecutive passes end the game. A draw is worth 0.5 points.
Scoring uses strict Tromp-Taylor area scoring with no dead stone removal.

Coordinates: columns A B C D E F G H J from left to right (I is skipped); rows 9 through 1 from
top to bottom. B = Black, W = White, Z = empty.

ABCDEFGHIJ
9 ZZZZZZZZ
8 ZZZZZZZZ
7 ZZZZZWZZ
6 ZZZZZZZZ
5 ZZZZBZZZ
4 ZZZZZZZZ
3 ZZZZZZZZ
2 ZZZZZZZZ
1 ZZZZZZZZ

Recent 5 moves (oldest to newest): B E5, W G7

Currently illegal because of ko/superko: none

Legal moves: A1, B1, C1, D1, E1, F1, G1, H1, J1, A2, B2, C2, D2, E2, F2, G2, H2, J2, A3, B3,
C3, D3, E3, F3, G3, H3, J3, A4, B4, C4, D4, E4, F4, G4, H4, J4, A5, B5, C5, D5, F5, G5, H5, J5,
A6, B6, C6, D6, E6, F6, G6, H6, J6, A7, B7, C7, D7, E7, F7, H7, J7, A8, B8, C8, D8, E8, F8, G8,
H8, J8, A9, B9, C9, D9, E9, F9, G9, H9, J9, pass, resign

Legal moves is authoritative for the current position and already accounts for ko and
positional superko.

Your entire final answer must be one entry copied exactly from Legal moves.
Write that entry on a single line, then end the response immediately.
```

3 Experimental results

3.1 Track 1: LLMs without tools

We evaluate 18 multi-turn API configurations and one single-turn baseline without tool calls (Tables 1 and 2). Figure 1 shows the multi-turn configurations. Rank ranges take into account overlapping confidence intervals.

GPT-6 Astra at max leads the Track 1 comparison at 2568 ± 140 Elo and \$0.154 per move. It is 764 Elo below the fast policy-only KataGo reference while costing approximately 1.1 million times more per move. The strongest KataGo player is 1,859 Elo higher and costs about 31 times less per move.

Table 1: Multi-turn API evaluation results and two KataGo references. Illegal moves are counts divided by played moves.

Rank	Player (effort)	Elo	95% CI	Games	Illegal moves	Seconds / move	Cost / move
1	GPT-6 Astra (max)	2568	± 140	30	0/1052	66	\$0.15
2–4	GPT-6 Astra (high)	2227	± 138	30	0/1015	10	\$0.033
2–6	Opus 5 (high)	2076	± 139	30	0/1239	26	\$0.090
2–8	GPT-5.6 Sol (max)	1929	± 207	14	4/516	79	\$0.081
3–10	GPT-5.6 Sol (high)	1846	± 207	14	24/566	27	\$0.034
3–11	Gemini 3.1 Pro (high)	1797	± 178	20	1/763	41	\$0.063
4–13	DeepSeek V4.1 Flash (max)	1707	± 206	14	1/565	140	\$0.022
4–13	Gemini 3.8 Flash (high)	1616	± 206	14	0/512	13	\$0.025
5–17	DeepSeek V4 Flash 0731 (max)	1449	± 227	14	160/603	160	\$0.016
5–17	Muse Spark 1.3 (high)	1449	± 227	14	21/504	46	\$0.0020
6–17	Muse Spark 1.3 (xhigh)	1415	± 217	14	19/596	38	\$0.0025
7–18	DeepSeek V4.1 Flash (high)	1380	± 205	14	4/692	70	\$0.012
7–18	Gemini 3.6 Flash (high)	1336	± 223	14	0/662	5.3	\$0.020
9–18	DeepSeek V4 Flash 0731 (high)	1189	± 197	14	3/544	150	\$0.014
9–18	GPT-5.6 Luna (high)	1091	± 190	14	3/806	17	\$0.0024
9–18	GPT-5.6 Luna (max)	1058	± 189	14	0/800	32	\$0.0043
9–18	Grok 4.6 (xhigh)	1044	± 189	14	9/500	35	\$0.050
12–18	Grok 4.6 (high)	916	± 263	14	19/551	30	\$0.044
–	KataGo (fast policy)	3332	± 32	4002	–	0.0072	$\$1.4 \times 10^{-7}$
–	KataGo (highest Elo)	4427	± 35	2992	–	250	\$0.0049

Single-turn versus multi-turn context. For GPT-6 Astra at high, multi-turn context reduces time by a factor of four and cost by 40% while maintaining Elo.

Table 2: GPT-6 Astra at high: single-turn versus multi-turn context. Both configurations have zero reported illegal moves and API problems. Cached input is the fraction of input tokens served from cache.

Context	Elo (95% CI)	Games	Cached input	Output tokens per move	Seconds / move	Cost / move
Single-turn	2011 ± 217	14	0.00%	1070.49	44.70	\$0.0581
Multi-turn	2227 ± 138	30	89.50%	196.61	10.32	\$0.0332

Correlation with ARC-AGI. As shown in Figure 3, LLMs’ Track 1 Elos strongly correlate with their ARC-AGI scores.

3.2 Track 2: LLMs with coding tools

In Track 2, the model performs continual learning for $N \in \{0, 1, 2, 4, 8\}$ hours and is then evaluated with the resulting context checkpoint. Each evaluation game starts from the same context checkpoint,

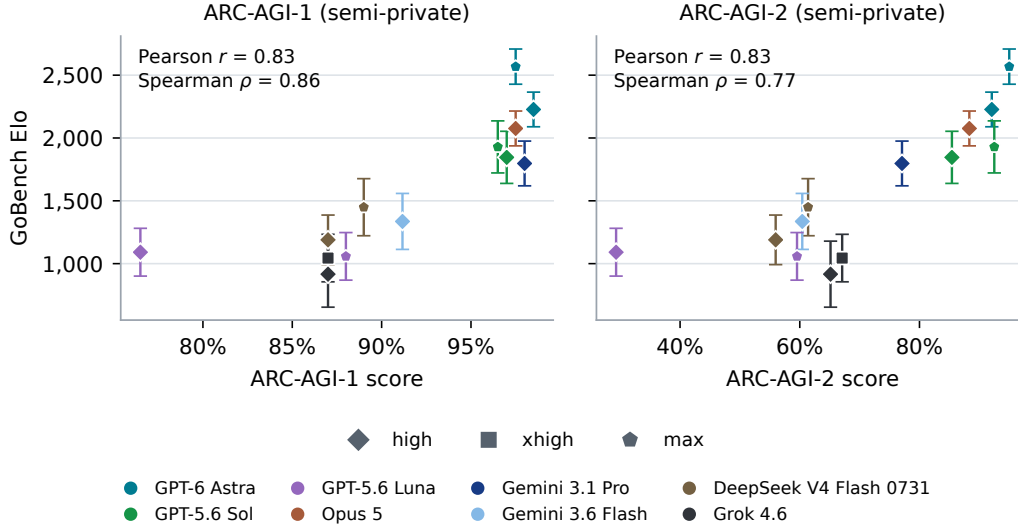


Figure 3: GoBench Elo versus ARC-AGI-1 and ARC-AGI-2 semi-private scores.

which does not include changes made during prior evaluation games. The model may further update the context during the evaluation game. Each evaluation game is capped at 30 minutes, and exceeding the cap counts as a loss. The remaining time is always available to the agent through a continuously updated `clock.json` file and in the turn prompt.

Continual learning can mean updating weights or context. In this work, we update the LLM’s context while keeping its weights fixed. Context can be thought of as files and directories in a file system, and it is synonymous with agentic harness state.

We evaluate Codex with GPT-6 Astra and GPT-5.6 Sol, both at `high`. We disable internet access and access to KataGo source code and results, while allowing other local tools. We use Bubblewrap as the sandbox and limit the compute resources to one physical CPU core, 4GB of RAM, and 2GB of disk. The Codex process is inside the sandbox, and it reaches the OpenAI API endpoint through a reverse proxy over a Unix domain socket. The socket messages are logged so that further investigation can be conducted to check for cheating, such as trying to reach the internet.

We considered giving the model access to KataGo during the continual learning phase, but we eventually decided against this idea because the model would just create an opening book by using the best KataGo model.

As shown in Figure 4, GPT-6 Astra in Codex with 0 hours of continual learning matches the best-performing model from Track 1. Its Elo improves with 1h and 2h of continual learning, while the 4h and 8h runs show no improvement beyond the 2h run.

GPT-5.6 Sol improves as continual learning duration increases from 0h to 1h, but it degrades as duration increases to 2h, 4h, 8h.

By using coding tools, they built C++ Go engines with Monte Carlo tree search, tactical rules, learned move-selection policies, and precomputed opening books, then used them to choose moves during evaluation.

The reverse proxy logs showed no signs of internet access attempts.

3.3 Manually inspecting the LLM games

Manually inspecting the LLM games revealed two potential concerns, positional superko and KataGo’s passing behavior, and we wrote a checker for each.

The Go Text Protocol (GTP) is the interface used for games between KataGo and LLMs, but the engine’s GTP `play` command accepted moves that violate positional superko. We implemented our

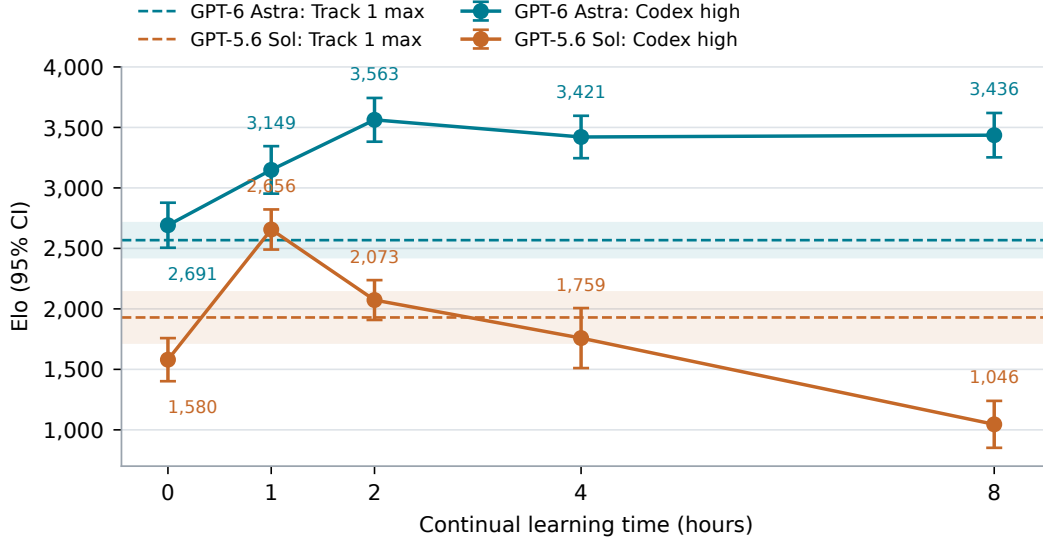


Figure 4: GPT-6 Astra and GPT-5.6 Sol at high in Codex after 0–8 hours of continual learning. Error bars show 95% confidence intervals. Dashed lines and shaded bands show each model’s Track 1 max baseline.

own checker to prevent illegal moves and reran the experiments after identifying violations in an earlier set of games.

Strict area scoring has no automatic dead stone removals, but KataGo players around 2000 Elos are very likely to pass while leaving opponent’s dead stones unremoved. This behaviour disappears as the KataGo players achieve over 3000 Elo, as shown in Figure 5.

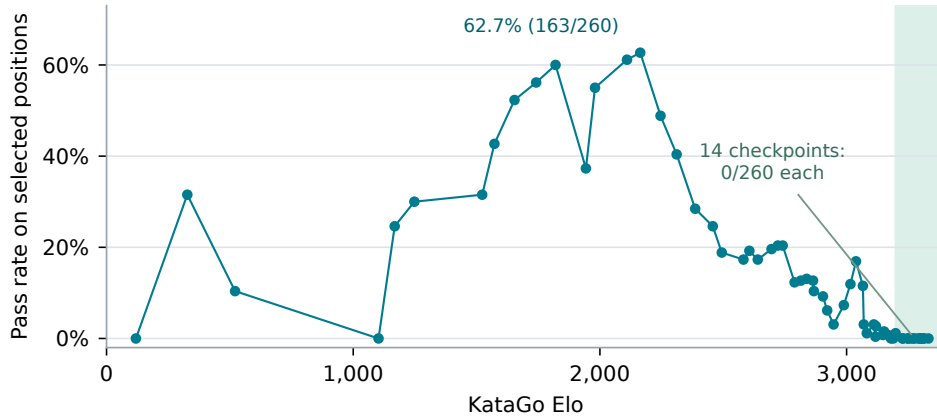


Figure 5: Pass rates on 13 historical positions across 63 KataGo b6c96 checkpoints.

4 Related work

Go has long served as a testbed for artificial intelligence. AlphaGo combined deep neural networks with tree search to achieve superhuman performance [Silver et al., 2016], while AlphaZero learned Go, chess, and shogi through self-play without expert demonstrations [Silver et al., 2018]. KataGo subsequently improved the efficiency of self-play training and is open source [Wu, 2019].

KataGo-Bench-1K [Ma et al., 2026] contains 1,000 board states and scores next-move predictions using KataGo as the judge. However, it does not involve complete games, providing only a myopic view of the players’ strengths.

Several active benchmarks evaluate LLMs through games and puzzles. Kaggle Game Arena, the AI Chess Leaderboard, and LLM CHESS evaluate performance on chess [Kaggle, 2026, Dubesor, 2026, Kolasani et al., 2025]; GTO Wizard Benchmark evaluates performance on poker [Provost et al., 2026]; and ARC-AGI-3 and MazeBench evaluate performance on puzzle games [ARC Prize Foundation, 2026a, Pappas and Pappas, 2026]. To our knowledge, GoBench is the first benchmark evaluating LLMs’ performance on complete games of Go, providing a novel addition to the set of benchmarks.

5 Directions for future work

Continual learning with weight updates. We experimented only with continual learning through context updates. Weight updates are necessary for Tracks 3 and 4, where the LLM performs continual learning with tools but is evaluated without tools. Track 4 matches the environment in which human professionals play Go, and it is essential for measuring LLMs’ learning efficiency relative to that of humans.

Adversarial robustness. Superhuman KataGo networks have been defeated by substantially weaker agents using adversarial strategies [Wang et al., 2023]. We can similarly study whether LLMs are susceptible to adversarial attacks in the game of Go, and whether they can be trained to achieve adversarial robustness.

Video game benchmarks. GoBench evaluates reasoning on text. Future benchmarks can further incorporate visual perception and physical control, providing a path toward physical intelligence.

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A Color function

We introduce a color function that estimates the Black advantage given the average Elo of the two players. We first fit a Bradley–Terry Elo model without a color function and denote its ratings by b_i . In the second stage, we fit the final player ratings and the color function while keeping b_i frozen.

We conduct an ablation study to find the best color function. To understand its generalization to unseen player matchups, we conduct the train-test split not by game, but by player. We randomly select half of the players, assign the matchups between them as the train set, and assign the rest of the games as the test set. Given 133 KataGo players and 300K games, we estimate b_i for all the players first, and then calculate mean train and test losses over 40 repetitions of train-test sampling. This ablation uses an earlier 300K-game dataset. The current historical rating pool is listed in Appendix D; both pools include the random anchor and kata1-zhizi-b40c768nbt-fdx6c.

The results are shown in Table 3. We mark a color function as statistically worse than the quadratic one when the 95% Student’s t interval over its 40 paired test-loss differences lies entirely above zero. A one-Elo knot spacing that overfits the train set generalizes poorly to the test set, as expected. We use the piecewise-linear color function for its simplicity; its test loss ties with the quadratic.

Table 3: Ablation study on the color function.

Color function	Color params	Train loss	Test loss	Worse than quadratic?
Piecewise linear (1,500 Elo)	4	0.3658	0.3664	No
Quadratic	3	0.3660	0.3664	No
Piecewise constant (500 Elo)	9	0.3657	0.3667	No
Negative exponential	2	0.3662	0.3668	Yes
Cubic spline	8	0.3653	0.3674	Yes
Piecewise linear (500 Elo)	10	0.3653	0.3676	Yes
Piecewise constant (250 Elo)	17	0.3652	0.3676	Yes
Piecewise linear (250 Elo)	18	0.3649	0.3676	Yes
Global color term	1	0.3683	0.3677	Yes
No color term	0	0.3709	0.3699	Yes
Piecewise linear (1 Elo)	4,270	0.3521	0.4894	Yes

Even though using the color function has negligible influence on the ranking of the players, it improves the predicted win probabilities, as shown by the better test loss in Table 3.

B Elo estimation and uncertainty

This appendix gives the statistical details of the two-stage Elo model described in Section 2. In the first stage, we fit a Bradley–Terry model without a color term to obtain baseline ratings b_i . These ratings are then frozen. In the second stage, we jointly fit the reported ratings r_i and the parameters of the Black-advantage function f .

B.1 Outcome model

If r_i is the Elo rating of player i , then the probability that player i wins as Black against player j is

$$p_{ij} = \sigma(c[r_i - r_j + f((b_i + b_j)/2)]), \quad c = \frac{\ln 10}{400}, \quad \sigma(x) = \frac{1}{1 + \exp(-x)},$$

where b_i and b_j are the frozen first-stage baseline ratings and f is the color function.

B.2 Maximum a posteriori estimation

Given a list of game results $G = [(i, j, y), \dots]$, where player i is Black, player j is White, and y is the game result in $\{1, 0, 1/2\}$, we estimate the Elo ratings by maximizing the regularized Bradley–Terry log-likelihood

$$\mathcal{L}(\theta) = \sum_{(i,j,y) \in G} [y \log p_{ij} + (1 - y) \log(1 - p_{ij})] - \frac{1}{2} \sum_k \frac{(\theta_k - \mu_k)^2}{s_k^2},$$

where θ includes both player ratings and Black-advantage parameters, and μ_k and s_k are the mean and standard deviation of the independent Gaussian prior on parameter k .

For the Gaussian prior, we use $N(0, 10000^2)$ for both KataGo player ratings and color-function parameters, but $N(1000, 2000^2)$ for LLMs to speed up their convergence.

B.3 Laplace confidence intervals

Uncertainty in the fitted parameters is approximated using a Laplace approximation [Tierney and Kadane, 1986] around the maximum a posteriori estimate $\hat{\theta}$. The information matrix at the fitted parameters is

$$\hat{\mathbf{I}} = c^2 \sum_g \hat{p}_g (1 - \hat{p}_g) \mathbf{x}_g \mathbf{x}_g^\top,$$

where $\mathbf{x}_g$ denotes the fixed second-stage feature vector for game g . Its player components are +1 for the Black player and −1 for the White player, and its color components are the piecewise-linear basis values evaluated at the frozen average baseline Elo.

Including the curvature of the independent Gaussian priors, the negative Hessian of the log posterior is

$$\hat{\mathbf{H}} = \hat{\mathbf{I}} + \text{diag}(s_1^{-2}, \dots, s_d^{-2}),$$

where d is the number of optimized parameters. The approximate posterior covariance matrix is $\mathbf{C} = \hat{\mathbf{H}}^{-1}$, and the marginal 95% confidence interval is

$$\hat{r}_i \pm 1.96 \sqrt{C_{ii}}.$$

These intervals are conditional on the frozen baseline ratings; they do not propagate uncertainty from the first-stage no-color fit.

The player with the random policy is fixed at 0 Elo as an anchor and is not optimized in θ or included in the covariance matrix.

B.4 Limitations of the Elo model

Elo ratings hide non-transitive effects, where a low-Elo player beats a high-Elo player with more than 50% probability. Displaying game outcomes between all players in a 2D matrix can be beneficial, but we stick to Elo for its simplicity.

C Matchmaking

We want to schedule games that shrink the confidence intervals as much as possible. To do this, we first introduce the gain matrix, where each entry (i, j) is defined as the predicted reduction in the sum of confidence intervals of players we care about:

$$G_{ij} = \sum_{q \in \mathcal{S}} \left[\sqrt{C_{qq}} - \sqrt{C_{qq}^{(ij)}} \right],$$

where $C^{(ij)}$ is the predicted covariance matrix after players i and j play a pair of color-swapped games, and $\mathcal{S}$ is the set of players we care about.

The games are scheduled in batches, and the games within a batch are played concurrently. Before each batch, the Elo ratings, confidence intervals, and gain matrices are recalculated using all completed games.

In the first stage, the KataGo players play a total of 320K games among themselves in batches of 4K. In each batch, 2K color-swapped pairs are sampled with probability proportional to the gain of each pair.

In the second stage, LLMs play against KataGo in batches of two color-swapped games. In total, LLMs play 420 games: 320 API games across 19 configurations (14–30 games each) and 100 Codex games across five continual-learning budgets (20 each). In each batch, the pair that maximizes the gain is chosen.

Why are the matchmaking algorithms different for the two stages? The ground-truth gain matrix is calculated only before every batch. Predicting the gain matrix after N games requires modeling up to 3^N possible outcomes, given 3 potential outcomes per game. Because predicting the gain matrix is intractable, we simply sample with probability proportional to the gain. As an added benefit, such sampling is more robust to the non-transitivity in the Elo model.

The final 95% confidence-interval half-widths are 16–36 Elo for non-anchor KataGo players and 138–263 Elo for LLMs.

D KataGo player pool

Table 4 lists every KataGo player used as an Elo anchor. Names of the form `bAcB-sN` identify a KataGo network with A residual blocks and B channels after N training steps. An `x2` suffix marks a network trained with two GPUs for gradients instead of one, and `nbt` marks the newer nested-bottleneck architecture. A `-pP` suffix denotes tree search with P playouts; players without one use `numPlayouts = 1`.

A `-temp-T` suffix changes the policy-sampling temperature schedule of Section 2, which by default decays from 0.5 to 0.1. For $T \neq 0.3$, the suffix pins the temperature to the constant T for the whole game. The single exception is $T = 0.3$, which keeps the decaying schedule but raises its floor, decaying from 0.5 to 0.3. The purpose is to make the low-Elo population denser.

Table 4: The 133 KataGo players in the historical rating pool, ordered by Elo. Player names omit the `kata1-` prefix and the `-dN` data-count segment; `random` is the uniform-random anchor fixed at 0 Elo. CI is the half-width of the 95% confidence interval. The Zhizi network lacks a timing measurement and is omitted only from the cost plot.

Player	Elo	CI	Player	Elo	CI	Player	Elo	CI
b28c512nbt-s8566598912-p600	4427	35	b6c96-s141089536	3302	32	b6c96-s16525312-temp-0.5	2160	27
b28c512nbt-s12313658112-p600	4385	35	b6c96-s130521856	3297	31	b6c96-s14649344	2110	29
b28c512nbt-s7229617920-p600	4384	35	b6c96-s115648256	3290	31	b6c96-s14649344-temp-0.3	2084	27
b18c384nbt-s9761732864-p600	4368	35	b6c96-s103950080	3275	31	b6c96-s16525312-temp-0.7	2011	27
b28c512nbt-s12313658112-p60	4346	34	b6c96-s108812288	3270	31	b6c96-s13733120-temp-0.3	1985	27
b28c512nbt-s8566598912-p60	4332	34	b6c96-s95425792	3260	31	b6c96-s13733120	1980	29
b28c512nbt-s7229617920-p60	4308	34	b6c96-s89508352	3249	31	b6c96-s12849664	1943	28
b18c384nbt-s9761732864-p60	4278	34	b6c96-s83588096	3248	31	b6c96-s13733120-temp-0.5	1924	26
zhizi-b40c768nbt-fdx6c	4216	36	b6c96-s73091584	3228	31	b6c96-s12849664-temp-0.5	1856	26
b28c512nbt-s12313658112	4199	34	b6c96-s78659584	3227	31	b6c96-s11888896	1820	26
b28c512nbt-s8566598912	4191	34	b6c96-s69427456	3198	31	b6c96-s11888896-temp-0.5	1742	26
b28c512nbt-s7229617920	4161	34	b6c96-s65341184	3189	31	b6c96-s10825472	1742	28
b18c384nbt-s9761732864	4131	34	b6c96-s59560704	3180	31	b6c96-s14649344-temp-0.9	1676	26
b60c320-s6678602752	4102	34	b6c96-s63439616	3168	31	b6c96-s10014464	1654	27
b40c256-s11840935168	4071	34	b6c96-s55682304	3152	31	b6c96-s10014464-temp-0.3	1619	25
b40c256-s8049900800	4063	34	b6c96-s56669696	3147	31	b6c96-s13733120-temp-0.9	1583	25
b40c256x2-s4120339456	4027	34	b6c96-s52864512	3119	31	b6c96-s8982784	1572	27
b40c256-s5793846528	3994	34	b6c96-s50894592	3117	31	b6c96-s8080640	1523	27
b40c256x2-s4833666560	3987	33	b6c96-s48921344	3110	31	b6c96-s8982784-temp-0.3	1519	25
b20c256x2-s5055114240	3931	34	b6c96-s46949632	3082	31	b6c96-s8080640-temp-0.3	1413	24
b20c256x2-s3354994176	3898	33	b6c96-s45189632	3070	30	b6c96-s8982784-temp-0.5	1364	24
b20c256x2-s2430231552	3861	33	b6c96-s43286272	3066	31	b6c96-s6127360	1248	26
b20c256x2-s1610809600	3847	33	b6c96-s41312768	3038	31	b6c96-s8080640-temp-0.5	1246	23
b20c256x2-s1039565568	3841	33	b6c96-s39339776	3016	30	b6c96-s5214720	1168	26
b20c256x2-s668214784	3797	33	b6c96-s37368064	2989	31	b6c96-s8982784-temp-0.7	1137	23
b15c192-s1062526208	3752	33	b6c96-s35356160	2947	30	b6c96-s4136960	1103	22
b15c192-s798345984	3750	33	b6c96-s33239552	2921	31	b6c96-s10014464-temp-0.9	1062	22
b20c256-s241116160	3736	33	b6c96-s32344832	2904	30	b6c96-s8080640-temp-0.7	996	22
b15c192-s449394432	3700	33	b6c96-s30394624	2867	31	b6c96-s4136960-temp-0.3	917	22
b15c192-s315428352	3689	33	b6c96-s31390976	2864	31	b6c96-s5214720-temp-0.5	859	21
b15c192-s230670592	3638	33	b6c96-s29422336	2838	30	b6c96-s4136960-temp-0.5	731	20
b15c192-s172540416	3636	33	b6c96-s28460288	2815	31	b6c96-s5214720-temp-0.7	693	20
b10c128-s501483520	3565	32	b6c96-s27415296	2789	30	b6c96-s6127360-temp-0.9	640	19
b10c128-s356821760	3556	32	b6c96-s26363136	2741	31	b6c96-s4136960-temp-0.7	563	19
b10c128-s182492416	3535	32	b6c96-s25461504	2721	30	b6c96-s1995008	521	22
b10c128-s258941696	3529	32	b6c96-s24455424	2696	31	b6c96-s4136960-temp-0.9	440	17
b10c128-s146897408	3508	32	b6c96-s23401472	2640	29	b6c96-s1995008-temp-0.3	328	16
b10c128-s108710656	3494	32	b6c96-s22371328	2606	31	b6c96-s938496	327	21
b10c128-s82632704	3459	32	b6c96-s21371392	2582	29	b6c96-s938496-temp-0.3	235	16
b10c128-s56992512	3438	32	b6c96-s20302336	2494	31	b6c96-s1995008-temp-0.9	168	16
b10c128-s41138688	3388	32	b6c96-s19408128	2458	29	b6c96-s1248000	119	20
b6c96-s175395328	3332	32	b6c96-s18429184	2386	29	b6c96-s1248000-temp-0.3	113	16
b6c96-s152505856	3314	32	b6c96-s17434112	2311	28	random	0	0
b6c96-s165180416	3312	31	b6c96-s16525312	2246	28			
b6c96-s122554112	3304	32	b6c96-s15704832	2164	29			